

Problem-Statement Decision Brief

Space & Aerospace · FAR AWAY 2026

Optimised for: builder-first hackathon scoring + maximum resume / recruiter pull

CHOSEN PROJECT

OrbitGuard explainable satellite collision screening + 3D globe

Maximum judge wow (live 3D globe). Standout resume line. 100% software + public data.
Validated against the official TracSS answer key (Jan 2026) — real, defensible recall numbers.

Claude's overall rating

8.0 / 10

Software-only

Resume-first

Team of 4-5

OrbitGuard — project brief

This brief covers OrbitGuard, the chosen project for FAR AWAY 2026, researched against the live web in June 2026. The project is rated by Claude on six axes (1-5), given an overall out of 10, and carries an explicit STAR NOVELTY — the specific edge that makes it stand out. Theme: Space & Aerospace.

Why OrbitGuard: maximum judge wow (live 3D globe of real orbits + collision alerts), standout resume line (aerospace + ML + visualisation), 100 percent software + 100 percent public data, strong India angle (Pixxel / GalaxEye / ISRO), and the TraCSS answer key (Jan 2026) gives real, defensible accuracy numbers — not a toy globe.

Scoring legend

Axis	What it measures
Resume / recruiter pull	How strongly it makes a recruiter want to interview you
Real-world impact	Does it solve a real, felt problem? Could it be sold / shipped?
Technical depth	Real engineering vs a thin 'AI wrapper' (which the rubric penalises)
Originality	How crowded the space already is - how novel your angle is
Demo legibility	Can a non-technical judge instantly 'get' the 2-minute demo?
Buildability	Realistic to ship a strong version with a 4-5 person team in the window

OrbitGuard at a glance

Problem statement	Res.	Impact	Tech	Orig.	Demo	Build	Overall
OrbitGuard Space and Aerospace	4	3.5	4	4	5	3.5	8.0

Cell colour = score band (green high to red low). Deep-dive profiles follow, one per page.

THE DECISION

- **Chosen project:** OrbitGuard — Space & Aerospace.
- **Why:** maximum judge wow (live 3D globe + collision alerts), unusual resume line, 100% software + public data.
- **Your edge:** explainability + small/Indian operators + TraCSS validation — not raw screening (SpaceX Stargaze does that).
- **Day-zero:** request TraCSS access, create Space-Track account, cache Celestrak TLEs, pick one primary satellite.

OrbitGuard

Explainable satellite collision-risk screening + 3D globe for small operators

CLAUDE'S RATING

8.0 / 10

CHOSEN PROJECT · #1 Theme: Space and Aerospace

★ **STAR NOVELTY (our edge):** Bench-marked space safety for the rest of us: a free browser 3D conjunction screen + an EXPLAINABLE ML risk score for small / Indian operators, validated against the Jan-2026 TraCSS OPEN answer-key dataset - real, defensible accuracy numbers, not a toy globe.

THE PROBLEM (plain English)

Small / university / Indian satellite operators get cryptic collision warnings and cannot afford the commercial tools (lakhs per year) to interpret them - so they fly blind or dodge needlessly. Until Jan 2026 there was not even open test data to build against.

HOW WE'D SOLVE IT

A free, browser-based conjunction-screening dashboard: public orbital data (TLEs), propagate orbits, screen for close approaches, an explainable ML risk score (probability + why), a 3D globe with a clear DODGE / WAIT / WATCH recommendation - validated against the new TraCSS answer key.

RATINGS

Resume / recruiter pull		Real-world impact	
Technical depth		Originality	
Demo legibility		Buildability	

PROS

- Huge judge wow (live 3D globe of real orbits + collision alerts).
- Unusual, memorable resume line (aerospace + ML + viz) that stands out in a CS pile.
- Genuinely fresh enabler: the TraCSS answer-key dataset only opened Jan 2026.
- 100% software, 100% public data; strong India angle (Pixxel / GalaxEye / ISRO).

CONS / RISKS

- Less of a sellable product / narrower user base than the exam ideas.
- SpaceX Stargaze + commercial SSA exist - your wedge is explainability + small operators.
- Orbital-mechanics correctness is unforgiving; the math must be right.

REAL-WORLD IMPACT & SELLABILITY Democratizes collision safety for operators priced out of commercial SSA; civic + space-policy resonance, especially in India.

LIMITATIONS & RISKS

- Niche user base; monetization unclear.
- Needs careful propagation (Skyfield) to be credible.
- Public TLEs are lower-accuracy than operator ephemerides.

HACKATHON SCOPE (MVP) A region of LEO + a few public satellites. TLEs (Celestrak / Space-Track), Skyfield propagation, close-approach screening, an ML / analytic risk score + natural-language explanation, a Cesium / Three.js 3D globe + DODGE / WAIT panel; report accuracy vs the TraCSS answer key.

TECH STACK Python (Skyfield) propagation + screening; a light ML risk model; claude-opus-4-8 for the 'why + recommendation'; Cesium / Three.js + Next.js; TraCSS + Celestrak public data.

FAR AWAY FIT + RESUME SIGNAL FAR AWAY: tops Demo / Innovation + a rare Space entry; mid Impact-sellability. Resume: 'explainable satellite-collision dashboard validated on the official TraCSS dataset' = standout, hard-to-fake aerospace + ML line.

OrbitGuard — build commitment

BUILD THIS: OrbitGuard — Space & Aerospace.

Free, explainable satellite collision-risk screening + 3D globe for small operators, validated against the official TraCSS answer key (Jan 2026). Maximum judge wow: a live 3D globe of real orbits with collision alerts. Standout, hard-to-fake resume line (aerospace + ML + visualisation). 100 percent software + 100 percent public data. Strong India angle (Pixxel / GalaxEye / ISRO).

DAY-ZERO CHECKLIST — DO BEFORE WRITING ANY CODE

- **1.** Request TraCSS dataset access (Google account, Office of Space Commerce) — NOT instant, do this first.
- **2.** Create a free Space-Track.org account to get CDM covariance data.
- **3.** Cache a CelesTrak GP TLE/OMM snapshot locally (do NOT hammer the API — rate-limit risk).
- **4.** Choose one primary satellite to anchor the MVP (e.g. Pixxel Firefly, GalaxEye SAR-1, or ISRO TechSat).
- **5.** Assign the five team lanes: propagation / screening+validation / risk+LLM / 3D globe / backend.
- **6.** Use OMM / JSON format for new objects (NORAD 5-digit IDs run out ~12 July 2026).

Key sources (June 2026): CelesTrak GP (TLE / OMM, free, 2-hour refresh); Space-Track.org (CDMs, free account); Skyfield (Python SGP4); TraCSS 'Dataset for Conjunction Assessment Verification' (Office of Space Commerce, Jan 2026, spherical + SFSH volume answer keys); Cesium ion / CesiumJS (3D globe, free tier); Three.js; Next.js; claude-opus-4-8 (LLM explainer).